Module 15) Advance Python Programming

2. Reading Data from Keyboard

- **Using the input() function to read user input from the keyboard.**

input() Function in Python – Theory Overview

The input() function in Python is used to **read user input from the keyboard** during program execution. It pauses the program and waits for the user to type something and press Enter.

Basic Behavior:

- The input() function always returns the **user input as a string**.
- It can optionally display a **prompt message** to guide the user.

Syntax:

input(prompt)

Key Characteristics:

1. **Returns a String**
   No matter what the user types, input() returns it as a string. You must **convert** it if another type is needed.
2. **Can Be Assigned to Variables**
   The value entered can be stored in a variable for later use.

3. **Used in Interactive Programs**
   Ideal for making your program interactive and dynamic.

- ## **Converting user input into different data types (e.g., int, float, etc.)**.

Converting User Input into Different Data Types in Python – Theory

In Python, the input() function **always returns user input as a string**, regardless of what the user types. To use this input in mathematical operations or other type-specific logic, it must be **explicitly converted** to the required data type.

Why Conversion is Needed:

- User input: "25" (a string)
- You cannot perform arithmetic like input + 5 unless it's converted to a numeric type like int or float.

Key Points:

1. **int()**: Converts strings that contain whole numbers.
   - Only valid for strings like "5", not "5.5" or "five".
2. **float()**: Converts strings with decimal numbers.
   - Accepts "5.0" or "3.14", but not "three point one".
3. **bool()**: Converts based on truthy or falsy values.
   - bool("") → False
   - bool("anything") → True
4. **Invalid conversions raise errors**
   - int("abc") or float("hello") will cause a ValueError.